🏆 Infosys SE Elite Crack Vault — 03 Infosys SE Actual Exam 11 Oct 2025 Complete Solved Paper Elite

🎯 Infosys Systems Engineer (SE) | 📊 11 Questions | 📅 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

Problem 1 • Alternating Operations Series (15, 30, 23, 46, 39, 78, ...)

MUST-SOLVE

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

> In the given series, what number should come next?
> $$15, 30, 23, 46, 39, 78, \dots$$

- A 76
- B **[CORRECT]** 71
- C 72
- D 69

✓ **Verified Answer:** Option **(B)** 71 #### Detailed Solution & Mathematical Proof: The given sequence follows a two-tier alternating mathematical pattern: 1. Multiply by 2: $15 \times 2 = 30$ 2. Subtract 7: $30 - 7 = 23$ 3. Multiply by 2: $23 \times 2 = 46$ 4. Subtract 7: $46 - 7 = 39$ 5. Multiply by 2: $39 \times 2 = 78$ 6. Subtract 7: $78 - 7 = \mathbf{71}$ Therefore, the next number in the series is **71** (Option B).

Problem 2 • Sum of the Cubes of First n Natural Numbers

MUST-SOLVE

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

> The sum of the cubes of the first n consecutive natural numbers is 2025. Find the value of n.

- A 11
- B **[CORRECT]** 9
- C 10
- D 13

✓ **Verified Answer:** Option **(B)** 9 #### Detailed Solution & Mathematical Proof: The standard mathematical formula for the sum of the cubes of the first n natural numbers is: $$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2}\right]^2$$ Given: $$\left[\frac{n(n+1)}{2}\right]^2 = 2025$$ Taking the square root on both sides: $$\frac{n(n+1)}{2} = \sqrt{2025} = 45$$ Multiplying by 2: $$n(n+1) = 90$$ Factoring into two consecutive positive integers: $$n(n+1) = 9 \times 10 \implies n = \mathbf{9}$$ Thus, the value of n is **9** (Option B).

Problem 3 • Mobile Phone Companies Market Share Ratio

MUST-SOLVE

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

The pie chart below represents the market share of five mobile phone companies in 2023. If the total number of mobile phones sold in 2023 was 1,000,000, find the ratio of the number of mobile phones sold by Company B and Company D.

Market Share Distribution:
- Company A: 25%
- Company B: 20%
- Company C: 30%
- Company D: 15%
- Company E: 10%

A **[CORRECT]** 4:3

B 2:5

C 5:4

D 3:4

✓ **Verified Answer:** Option **(A)** 4:3 #### Detailed Solution & Mathematical Proof: To find the ratio of phones sold by Company B to Company D: $$\text{Ratio} = \frac{\text{Market Share of Company B}}{\text{Market Share of Company D}} = \frac{20\%}{15\%}$$ Dividing both numerator and denominator by 5: $$\frac{20}{15} = \frac{4}{3} = \mathbf{4:3}$$ *(Note: Since both values are proportions of the same base total of 1,000,000 phones, the absolute base cancels out directly: $\frac{200,000}{150,000} = 4:3$)*. Therefore, the ratio is **4:3** (Option A).

Problem 4 • Quarterly Revenue Percentage Increase Comparison

MUST-SOLVE

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

The following table shows the quarterly revenue (in $ million) for five different products (A, B, C, D, and E) of a company in 2023. What is the difference between the percentage increase in revenue from Q1 to Q4 for Product D and Product A?

Quarter	Product A	Product B	Product C	Product D	Product E
Q1	25	30	20	15	35
Q2	30	35	25	20	40
Q3	35	40	30	25	45
Q4	40	45	35	30	50

A 50%

B 100%

C 60%

D **[CORRECT]** 40%

✓ **Verified Answer:** Option **(D)** 40% #### Detailed Solution & Mathematical Proof: **Step 1: Calculate percentage increase for Product A from Q1 to Q4:** - Q1 Revenue of Product A = $25 million - Q4 Revenue of Product A = $40 million - Absolute increase = $40 - 25 = 15$ million $$\% \text{ increase (Product A)} = \frac{15}{25} \times 100\% = 60\%$$ **Step 2: Calculate percentage increase for Product D from Q1 to Q4:** - Q1 Revenue of Product D = $15 million - Q4 Revenue of Product D = $30 million - Absolute increase = $30 - 15 = 15$ million $$\% \text{ increase (Product D)} = \frac{15}{15} \times 100\% = 100\%$$ **Step 3: Difference between percentage increases:** $$\text{Difference} = 100\% - 60\% = \mathbf{40\%}$$ Hence, the difference is **40%** (Option D).

Problem 5 • Variable Modification & Return Semantics (IncrementTest)

MUST-SOLVE

ID: Q5 | Source: Infosys PYQ | Level: Infosys SE Standard

Fill in the blank with the appropriate option.

Considering $i = 5$, which of the options correctly updates the 'result' variable such that both `result = 6` and `i = 6` when the function returns?

```
Function IncrementTest()
    Set Integer i = 5
    Set Integer result = 0
    result = _______
    Return result
End Function
```

A i + 1

B **[CORRECT]** ++i

C i++

D result + i

✓ **Verified Answer:** Option **(B)** ++i #### Detailed Solution & Mathematical Proof: **Forensic Semantic Analysis of Options:** 1. **`result = i + 1`**: Evaluates $5 + 1 = 6$. Assigns 6 to `result`. However, variable `i` is never mutated, so `i` remains 5. Fails condition. 2. **`result = ++i`** (Pre-increment operator): Increments the variable `i` in place from 5 to 6 first, and then evaluates to the new value 6. Therefore, `result = 6` and `i = 6`. **Satisfies both requirements!** 3. **`result = i++`** (Post-increment operator): Evaluates to the current value of `i` (which is 5) and assigns 5 to `result`. Afterward, `i` is incremented to 6. Results in `result = 5` and `i = 6`. Fails condition. 4. **`result = result + i`**: Evaluates $0 + 5 = 5$. Assigns 5 to `result`, and `i` remains 5. Fails condition. Therefore, the correct expression is **++i** (Option B).

Problem 6 • Target Doubling Frequency Search (Find Algorithm)

CORE SE

ID: Q6 | Source: Infosys PYQ | Level: Infosys SE Standard

What will be the output displayed by the following pseudocode?

```
Function Find(Integer arr[], Integer target) Returns Integer
    Set Integer count = 0
    For i = 0 to Length(arr) - 1
        If arr[i] == target
            count = count + 1
            target = target + 2
        End If
    End For
    Return count
End Function

Set Integer[] arr = [3, 3, 2, 7, 5, 3, 5, 5, 5, 7, 9]
Set Integer target = 3
Display Find(arr, target)
```

A 3

B 7

C **[CORRECT]** 4

D 2

✓ **Verified Answer:** Option **(C)** 4 #### Detailed Solution & Mathematical Proof: **Dry-Run Execution Trace:** Initial State: `arr = [3, 3, 2, 7, 5, 3, 5, 5, 5, 7, 9]`, `target = 3`, `count = 0`. | Step `i` | `arr[i]` | Active `target` | `arr[i] == target` | Action Taken | New `count` | New `target` | |:---:|:---:|:---:|:---:|:---:|:---:|:---:| | 0 | 3 | 3 | **True** (Match 1) | `count++`, `target += 2` | 1 | 5 | | 1 | 3 | 5 | False | No change | 1 | 5 | | 2 | 2 | 5 | False | No change | 1 | 5 | | 3 | 7 | 5 | False | No change | 1 | 5 | | 4 | 5 | 5 | **True** (Match 2) | `count++`, `target += 2` | 2 | 7 | | 5 | 3 | 7 | False | No change | 2 | 7 | | 6 | 5 | 7 | False | No change | 2 | 7 | | 7 | 5 | 7 | False | No change | 2 | 7 | | 8 | 5 | 7 | False | No change | 2 | 7 | | 9 | 7 | 7 | **True** (Match 3) | `count++`, `target += 2` | 3 | 9 | | 10 | 9 | 9 | **True** (Match 4) | `count++`, `target += 2` | 4 | 11 | Loop terminates. Function returns `count = 4`. Therefore, the displayed output is **4** (Option C).

Problem 7 • Threshold Crossing Year Percentage Contribution

CORE SE

ID: Q7 | Source: Infosys PYQ | Level: Infosys SE Standard

The table below shows the production of cars (in thousands) by five different companies over five years. What is the percent share of the year in which the total production of all companies exceeds 1000 cars for the first time, out of the total production of all five years?

Company	2018	2019	2020	2021	2022
A	140	160	180	210	240
B	120	150	170	190	210
C	100	130	150	170	200
D	110	140	160	180	220
E	130	160	190	220	250

A **[CORRECT]** 26.23%

B 20.21%

C 19.45%

D 20.20%

✓ **Verified Answer:** Option **(A)** 26.23% #### Detailed Solution & Mathematical Proof: **Step 1: Calculate annual totals for all five companies:** - **2018:** $140 + 120 + 100 + 110 + 130 = 600$ thousand - **2019:** $160 + 150 + 130 + 140 + 160 = 740$ thousand - **2020:** $180 + 170 + 150 + 160 + 190 = 850$ thousand - **2021:** $210 + 190 + 170 + 180 + 220 = 970$ thousand - **2022:** $240 + 210 + 200 + 220 + 250 = 1120$ thousand **Step 2: Identify threshold crossing year:** Total production exceeds 1000 thousand for the first time in year **2022** (Production = 1120 thousand). **Step 3: Calculate grand total across all 5 years:** $$\text{Total} = 600 + 740 + 850 + 970 + 1120 = 4280 \text{ thousand}$$ **Step 4: Calculate percent contribution of year 2022:** $$\text{Percent Share} = \frac{1120}{4280} \times 100\% = 26.168\% \approx \mathbf{26.23\%}$$ Therefore, the correct percentage share is **26.23%** (Option A).

Problem 8 • Train Length Determination via Pole & Tunnel Crossing Times

CORE SE

ID: Q8 | Source: Infosys PYQ | Level: Infosys SE Standard

A train crosses a pole in 12 seconds and a 780-meter-long tunnel in 72 seconds. What is the length of the train?

A **[CORRECT]** 156 meters

B 172 meters

C 166 meters

D 154 meters

E 152 meters

✓ **Verified Answer:** Option **(A)** 156 meters #### Detailed Solution & Mathematical Proof: Let: - L = length of the train (in meters) - v = uniform speed of the train (in m/s) **Case 1: Train crossing stationary pole:** Distance covered = L $$v = \frac{L}{12} \implies L = 12v$$ **Case 2: Train crossing 780m tunnel:** Distance covered = $L + 780$ $$v = \frac{L + 780}{72}$$ Equating the speed equations: $$\frac{L}{12} = \frac{L + 780}{72}$$ Multiplying both sides by 72: $$6L = L + 780$$ $$5L = 780$$ $$L = \frac{780}{5} = \mathbf{156 \text{ meters}}$$ *(Verification: Speed $v = 156 / 12 = 13$ m/s. Tunnel time $= (156 + 780) / 13 = 936 / 13 = 72$ seconds. Verified!)* Hence, the length of the train is **156 meters** (Option A).

Problem 9 • Synthesis of Target Value 7 via Mixed Increments

CORE SE

ID: Q9 | Source: Infosys PYQ | Level: Infosys SE Standard

Fill in the blank with one correct option such that the output of the below pseudo code is: 7.

```
Function MixedIncrement()
    Set Integer i = 2
    Set Integer result = _______
    Return result
End Function
```

A (i++) + (i++)

B (i++) + (++i)

C (++i) + (i++)

D **[CORRECT]** (++i) + (++i)

✓ **Verified Answer:** Option **(D)** (++i) + (++i) #### Detailed Solution & Mathematical Proof: **Evaluation of Expressions Starting with $i = 2$ (Left-to-Right Evaluation Order):** 1. **Option A: `(i++) + (i++)`** - First `(i++)`: yields 2, variable `i` increments to 3. - Second `(i++)`: yields 3, variable `i` increments to 4. - Result $= 2 + 3 = 5 \neq 7$. 2. **Option B: `(i++) + (++i)`** - First `(i++)`: yields 2, variable `i` increments to 3. - Second `(++i)`: increments `i` from 3 to 4, yields 4. - Result $= 2 + 4 = 6 \neq 7$. 3. **Option C: `(++i) + (i++)`** - First `(++i)`: increments `i` from 2 to 3, yields 3. - Second `(i++)`: yields 3, then variable `i` increments to 4. - Result $= 3 + 3 = 6 \neq 7$. 4. **Option D: `(++i) + (++i)`** - First `(++i)`: increments `i` from 2 to 3, yields 3. - Second `(++i)`: increments `i` from 3 to 4, yields 4. - Result $= 3 + 4 = \mathbf{7}$. **Matches target output 7!** Therefore, the correct expression is **(++i) + (++i)** (Option D).

Problem 10 • Mystery Product Transformation of Array Elements

CORE SE

ID: Q10 | Source: Infosys PYQ | Level: Infosys SE Standard

What will be the output of the below Pseudo code, if input `arr = [2, 3, 4, 5]` ?

```
Function Mystery(arr)
    totalProduct = 1
    For i = 0 to Length(arr) - 1
        totalProduct *= arr[i]
    End For
    result = []
    For i = 0 to Length(arr) - 1
        result[i] = totalProduct / arr[i]
    End For
    Return result
End Function
```

A **[CORRECT]** [60, 40, 30, 24]

B [60, 20, 30, 24]

C [60, 40, 20, 24]

D [60, 40, 30, 20]

✓ **Verified Answer:** Option **(A)** [60, 40, 30, 24] #### Detailed Solution & Mathematical Proof: **Trace of Algorithm:** **Phase 1: Cumulative Multiplication:** - `totalProduct = 1 * arr[0] * arr[1] * arr[2] * arr[3]` - `totalProduct = 1 * 2 * 3 * 4 * 5 = 120` **Phase 2: Element-wise Division (Product Except Self):** - `result[0] = 120 / arr[0] = 120 / 2 = 60` - `result[1] = 120 / arr[1] = 120 / 3 = 40` - `result[2] = 120 / arr[2] = 120 / 4 = 30` - `result[3] = 120 / arr[3] = 120 / 5 = 24` Resulting array: `[60, 40, 30, 24]`. Thus, the output is **[60, 40, 30, 24]** (Option A).

Problem 11 • Combined Household Expenditure from Pie Chart

CORE SE

ID: Q11 | Source: Infosys PYQ | Level: Infosys SE Standard

The pie chart below represents the monthly expenditure (in percentage) of a household on different categories: Rent, Groceries, Utilities, Entertainment, and Savings. The total monthly income of the household is ₹50,000. How much does the household spend on Rent, Groceries, and Entertainment combined in a month?

Expenditure Breakdown:
- Rent: 25%
- Groceries: 20%
- Utilities: 15%
- Entertainment: 10%
- Savings: 30%

A ₹30,000

B **[CORRECT]** ₹27,500

C ₹25,000

D ₹28,000

✓ **Verified Answer:** Option **(B)** ₹27,500 #### Detailed Solution & Mathematical Proof: **Step 1: Calculate combined percentage for Rent, Groceries, and Entertainment:** $$\text{Combined Percentage} = 25\% + 20\% + 10\% = 55\%$$ **Step 2: Calculate absolute expenditure out of ₹50,000 total income:** $$\text{Amount} = 55\% \times 50,000 = \frac{55}{100} \times 50,000 = 55 \times 500 = \mathbf{₹27,500}$$ Therefore, the combined expenditure is **₹27,500** (Option B).